AI-BASED EARLY MENTAL HEALTH BREAKDOWN DETECTION FROM SPEECH PATTERNS

Data Science and AI Lab Project (BSDA4001)
Milestone 3 – Model Architecture and Methodology

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PROBLEM STATEMENT

Objective

- Automatically detect depression risk and estimate PHQ-8 severity from multimodal clinical interview data (speech, language, facial behavior)

Why This Matters

- Depression affects 280 million people worldwide; most cases go undiagnosed
- Clinical assessment is subjective, infrequent, and resource-intensive
- Acoustic and linguistic speech patterns carry measurable, objective depression signals
- Early detection enables timely intervention before clinical breakdown occurs

Two tasks addressed:

- Binary risk classification - Healthy vs. Depressed (PHQ-8 threshold: 10)
- Continuous PHQ-8 severity regression - predicts score on 0–24 scale

SYSTEM OVERVIEW

Two Complementary Pipelines, One Clinical Task

Pipeline 1 - Classical ML (CPU, tabular features)

Pre-extracted features → 3-branch XGBoost classifiers → Logistic Regression meta-learner → Binary risk + PHQ-8 score

Pipeline 2 - Foundation Models (GPU, raw data)

Raw .wav audio → Frozen Whisper encoder → Audio embedding (512-D) Raw transcript → Frozen BioClinicalBERT → Text embedding (512-D) → Sigmoid Gate (element-wise fusion) → MLP Head → Binary risk + PHQ-8 score

Shared design principles:

- Both pipelines output the same two targets (binary label + PHQ-8 score)
- Class imbalance handled via weighted loss (pos_weight = 2.567) in both pipelines
- Three datasets: DAIC-WOZ (primary), RAVDESS (pre-training), MODMA (cross-corpus transfer)

DATASET ORGANIZATION

Three Datasets

Dataset	Purpose	Train/Val/Test	Features
DAIC-WOZ	Primary depression detection	107 / 35 / 47	448-D tabular
RAVDESS	Pipeline pre-training (8-class emotion)	840 / 300 / 300	338-D
MODMA	Zero-shot cross-corpus transfer	36 / 8 / 8	1,002-D

PREPROCESSING PIPELINE

PIPELINE 1 — TABULAR FEATURES

Modality	Input	Steps	Output Shape
Acoustic	COVAREP + formant CSVs	Session extraction → mean/std aggregation → imputation → branch split	(N, 323)
Linguistic	Transcript NLP features	Same pipeline; prefix filter nlp_	(N, 14)
Visual	OpenFace AUs, pose, gaze	Same pipeline; prefix filter au_, pose_, gaze_	(N, 111)

Pipeline 2 — Raw Modalities

- Audio: Resample 16 kHz → pad to 30 s → Whisper log-mel (80×3,000) → encoder (1500×768) → mean+std pool → (N, 1536) → linear → (N, 512)
- Text: BioClinicalBERT tokenizer → 512 tokens → [CLS] embedding (768-D) → linear → (N, 512)
- Imbalance: pos_weight = 77/30 = 2.567 applied in both pipelines via weighted loss

MODEL ARCHITECTURE

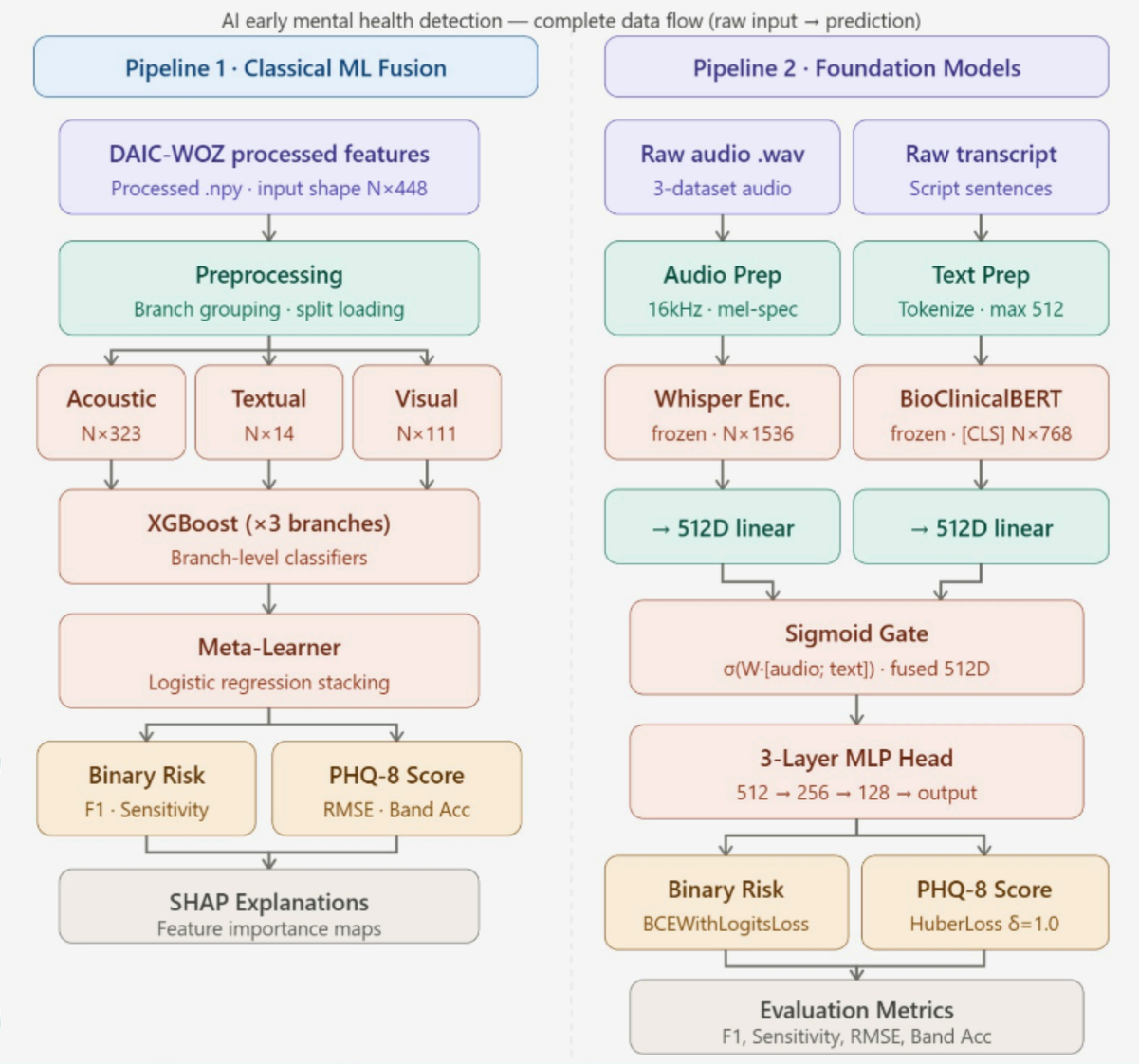
Pipeline 1 - XGBoost Stacking

- 3 independent XGBoost classifiers (one per modality branch)
- Soft probabilities → Logistic Regression meta-learner → Binary label
- Separate XGBoost regressor on all 448-D → PHQ-8 score
- Key params: n_estimators=300, max_depth=7, early_stopping=30

Pipeline 2 - GatedFusionHead

- Whisper (244M, frozen) + BioClinicalBERT (110M, frozen) → both projected to 512-D
- Sigmoid Gate: $g = \sigma(\text{Linear}(1024 \rightarrow 512))$ - element-wise 512-D vector, not scalar
- Fused = $g \odot \text{audio} + (1-g) \odot \text{text}$ → MLP: 512→256→128→{binary, PHQ-8}
- Trainable params: ~1.87M | Optimizer: AdamW | Scheduler: CosineAnnealingLR

DATA FLOW



1. **RAVDESS pre-training** - validates feature pipeline on 840 balanced samples; models discarded
2. **DAIC-WOZ depression detection** - 3 branches → meta-learner → binary + PHQ-8 output
3. **MODMA transfer** - zero-shot cross-corpus evaluation of the trained DAIC-WOZ model

INPUT-OUTPUT SPECIFICATIONS

Pipeline 1 : Input Shapes

- Full feature matrix - (N, 448) · float32
- Acoustic branch - (N, 323) · float32
- Linguistic branch - (N, 14) · float32
- Visual branch - (N, 111) · float32
- Meta-learner input - (N, 3) · float32 · stack of 3 branch probabilities
- Binary label - (N,) · int · threshold PHQ-8 ≥ 10
- PHQ-8 score - (N,) · float32 · continuous 0–24

Pipeline 2 : Tensor Shapes

- Raw audio - (480,000,) · float32 · 30s at 16kHz
- Whisper log-mel - (80, 3000) · float32
- Whisper encoder output - (1500, 768) · float32
- Audio pooled - (N, 1536) · float32 · mean+std across 1500 frames
- Audio projection - (N, 512) · float32
- Tokenized text - (N, 512) · int64 · padded to max 512 tokens
- BERT [CLS] embedding - (N, 768) · float32
- Text projection - (N, 512) · float32
- Fused representation - (N, 512) · float32 · after sigmoid gate
- Binary logit - (N, 1) · float32 · pre-sigmoid
- PHQ-8 prediction - (N, 1) · float32 · clipped to [0, 24]

ARCHITECTURE JUSTIFICATION

Why XGBoost in Pipeline 1

- Only 107 training samples, deep learning would overfit severely
- max_depth=7 + early_stopping=30 + L1/L2 regularization controls overfitting
- Naturally outputs calibrated probabilities needed by the meta-learner
- SHAP values give full feature-level explainability, a clinical requirement

Why Frozen Encoders in Pipeline 2

- Whisper trained on 680,000 hours of speech which cannot be reproduced from 107 samples
- BioClinicalBERT pretrained on clinical notes, domain-relevant out of the box
- Freezing reduces trainable parameters to ~1.87M, preventing overfitting
- Only the gate and MLP head learn task-specific patterns

ARCHITECTURE JUSTIFICATION

Why Sigmoid Gate over Concatenation

- Concatenation gives equal fixed weight to both modalities for every sample
- Element-wise 512-D gate learns per-dimension, per-sample weights
- Noisy audio → gate suppresses acoustic dimensions, amplifies text and vice versa

Strengths vs. Limitations

- Pipeline 1 - fast (~50–100 ms), interpretable, CPU-ready, but hand-crafted features
- Pipeline 2 - rich deep embeddings, temporal modeling, but slow (~500 ms–2 s), GPU required
- Both pipelines share the same clinical task and class-weighting strategy

END-TO-END PIPELINE VALIDATION

Pipeline 1 : Fully Verified on DAIC-WOZ

- All 27 required data files confirmed present before training
- All 3 branch models + meta-learner trained and saved to disk

Pipeline 2 : Integration Test

- Subset: 160 samples · train=120 · val=40 · 5 epochs · balanced classes
- Epoch 1 :Train Loss 2.6125 · Val Loss 2.0395 · Val F1 0.3333 · RMSE 5.5776
- Epoch 2 :Train Loss 2.1258 · Val Loss 2.0408 · Val F1 0.3333 · RMSE 5.4966
- Epoch 3 :Train Loss 2.1504 · Val Loss 2.0481 · Val F1 0.3333 · RMSE 5.0373
- Epoch 4 :Train Loss 2.0293 · Val Loss 2.0414 · Val F1 0.3333 · RMSE 5.2114
- Epoch 5 :Train Loss 2.0502 · Val Loss 2.0421 · Val F1 0.3333 · RMSE 5.3013
- Early stopping triggered at epoch 5

LOSS FUNCTIONS AND METRICS

Loss Functions

- Pipeline 1 Classification :Weighted Binary Cross-Entropy · pos_weight = 2.567 · eval metric: AUPRC
- Pipeline 1 Regression :Mean Squared Error
- Pipeline 2 Classification :BCEWithLogitsLoss · pos_weight = 2.567
- Pipeline 2 Regression :Huber Loss · delta = 1.0 · robust to PHQ-8 outliers at extreme scores
- Pipeline 2 Combined: L = 1.0 × L_clf + 0.3 × L_reg · regression down-weighted to stabilize classification

Validation Results - Classification (DAIC-WOZ)

- Acoustic - Macro F1: 0.60 · Sensitivity: 0.42 · ROC-AUC: 0.57
- Linguistic - Macro F1: 0.49 · Sensitivity: 0.25 · ROC-AUC: 0.49
- Visual - Macro F1: 0.43 · Sensitivity: 0.08 · ROC-AUC: 0.60
- Stacking Fusion - Macro F1: 0.39 · underperforms best branch

Validation Results - Regression (PHQ-8)

- RMSE: 6.56
- MAE: 5.48
- Within-1-Band Accuracy: 80% - clinically actionable

Why These Metrics

- Sensitivity is prioritized - missing a depressed patient is a higher-risk error than a false alarm
- AUPRC preferred over AUC-ROC for imbalanced classes
- Within-1-Band Accuracy reflects real clinical usability better than exact band match

CHALLENGES AND IMPROVEMENTS

Key Challenges

- Class imbalance with 30 depressed vs 77 healthy; sensitivity below 0.5 despite weighted loss
- Stacking fusion degrades performance, F1 drops from 0.60 (acoustic alone) to 0.39 (fused)
- Visual branch sensitivity = 0.08, static AU aggregates discard all temporal facial dynamics

Planned Improvements

- Replace logistic stacking with attention-based cross-modal fusion as designed in Pipeline 2
- Calibrate decision threshold to improve sensitivity at acceptable specificity cost
- Use MODMA for cross-corpus fine-tuning to augment the depressed class

CONCLUSION

What Was Built

- Two multimodal pipelines for depression detection from speech and clinical language
- Pipeline 1 : interpretable, CPU-deployable, SHAP-explainable, ~50–100 ms inference
- Pipeline 2 : foundation model architecture with dynamic gated fusion, ~1.87M trainable params

What Was Verified

- Pipeline 1 fully end-to-end verified on real DAIC-WOZ validation data, all models saved to disk
- Pipeline 2 integration-verified with tensor shapes, loss functions, and training loop confirmed correct

Core Findings

- Acoustic features are the strongest depression signal : Macro F1 = 0.60
- Simple modality fusion degrades when branches are correlated : stacking F1 drops to 0.39
- PHQ-8 regression achieves 80% within-1-band accuracy
- Sensitivity below 0.5 across all branches

THANK YOU